

Making Digestion Efficient

08/06/2026

Keywords: Hydrochloric acid, bile, stomach, liver, gall bladder, pancreas, neutralise, pH, surface area, emulsification

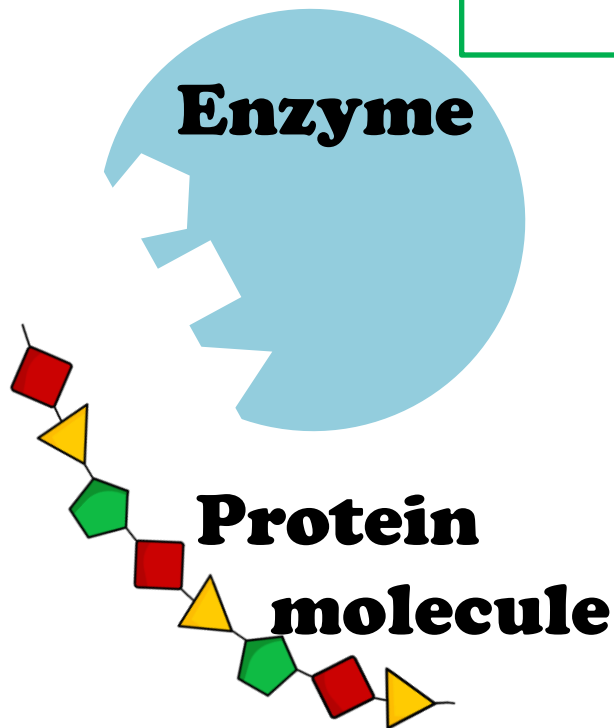
Starter: Answer these three review questions below.

EXT: What factors would affect how well the enzyme will work?

1. What is this enzyme called and where is it found?
[2 marks]

2. Name two more enzymes, where they are found and the substrate broken down [3]

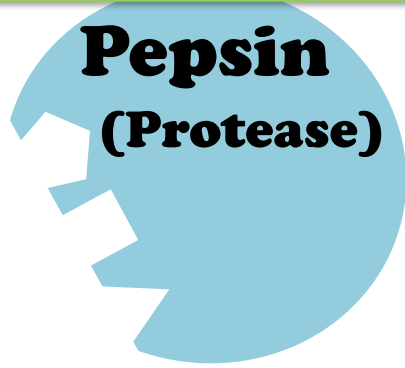
3. In the amylase pH required practical describe with examples all the variables. [4]



Review

1. What is this enzyme called and where is it found?

Pepsin
(Protease)



2. Name two more enzymes and where they are found and the substrate broken down

Amylase → **mouth** → **carbohydrate**

Lipase → **small intestine** → **fats/lipids**

Enzyme names [1] Locations [1] substrates [1]



Independent variable was the pH buffer



Dependent variable was the colour change

3. In the amylase pH required practical describe with examples all the variables. [4]



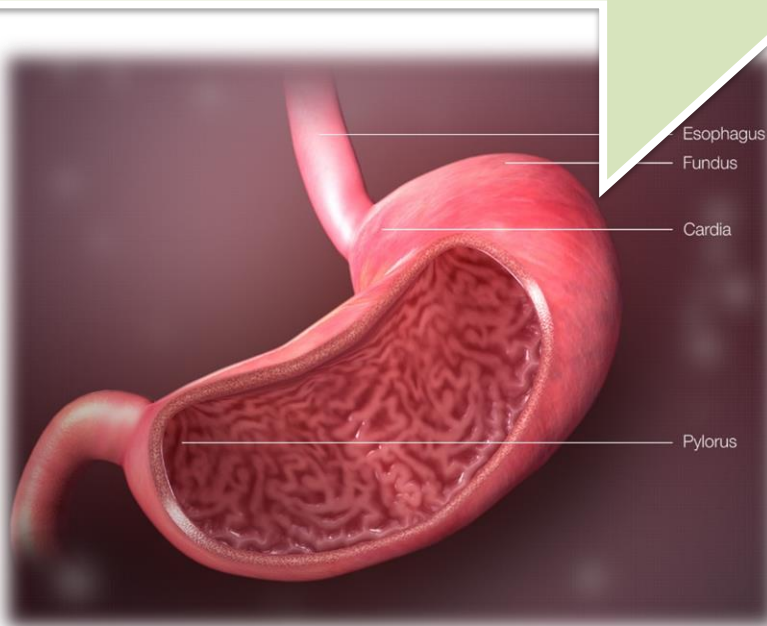
Control variable was temperature/ conc. of enzyme / concentration of starch

LOb: Understand the role of the stomach and liver in producing substances that increase efficiency of digestion.

Keywords: Carbohydrase, amylase, protease, amino acid, lipase, glycerol, fatty acid, soluble, absorb, emulsification

Learning Outcomes:

- Describe how the pH is changed in the digestive tract to improve efficiency
- Explain how surface area is altered to improve efficiency of digestion



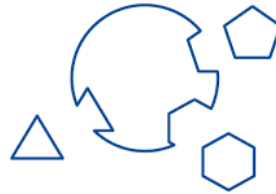
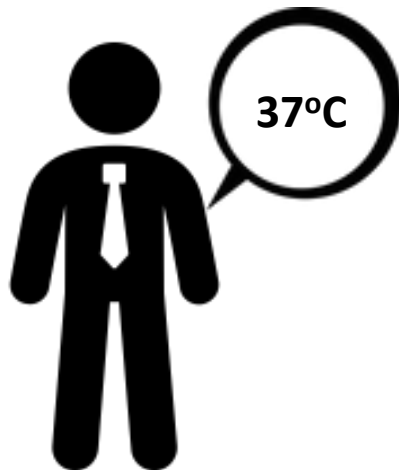
Did you know... the stomach has about 35 million glands in its lining. They secrete important substances to aid digestion.

Digestive enzymes

We've learnt that many different enzymes are produced by the digestive system.



Temperature is kept constant and every enzyme has the similar optimum temperature.



It is not as easy for pH as different enzymes have a different optimal

pH 2

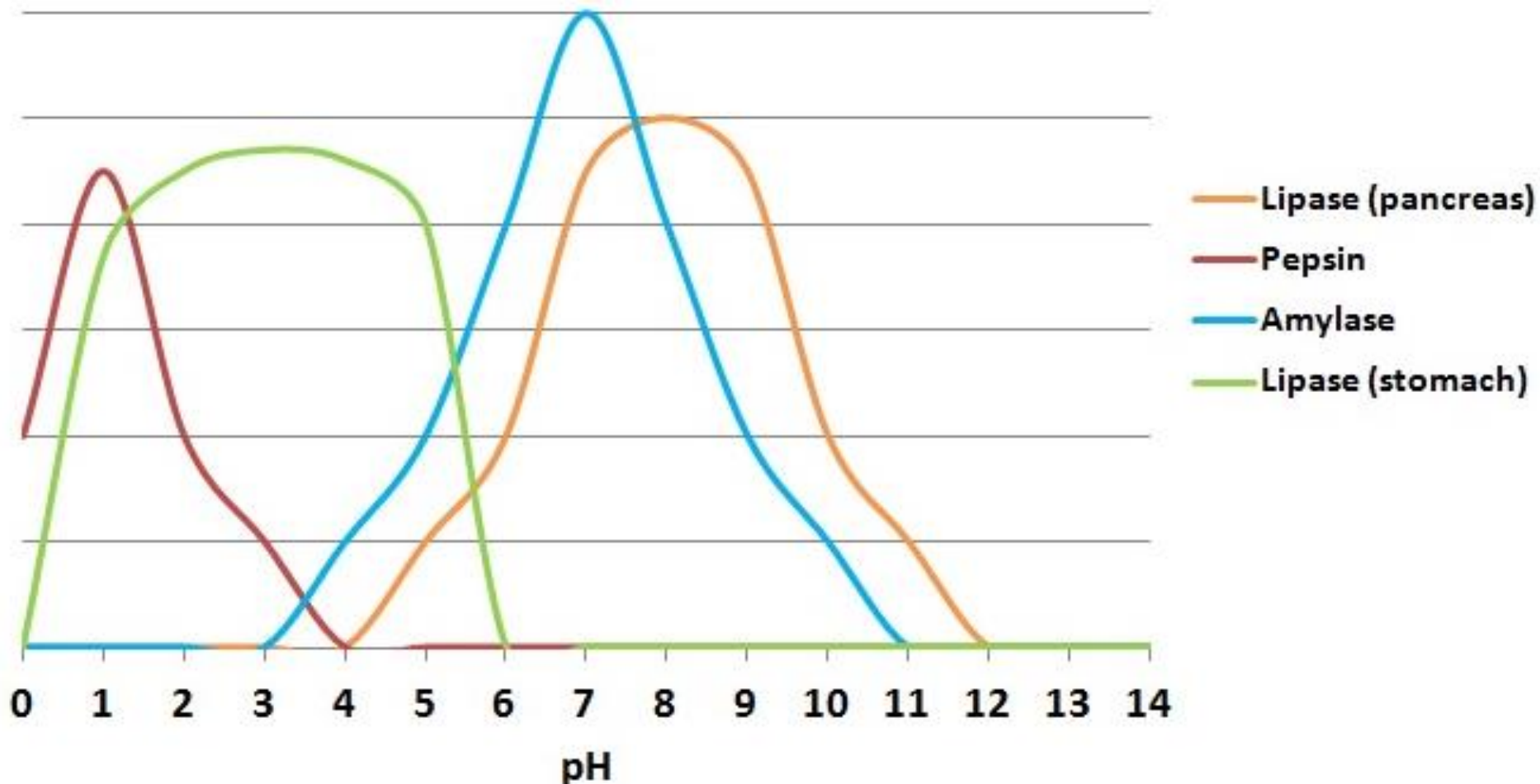
pH 7

pH 8

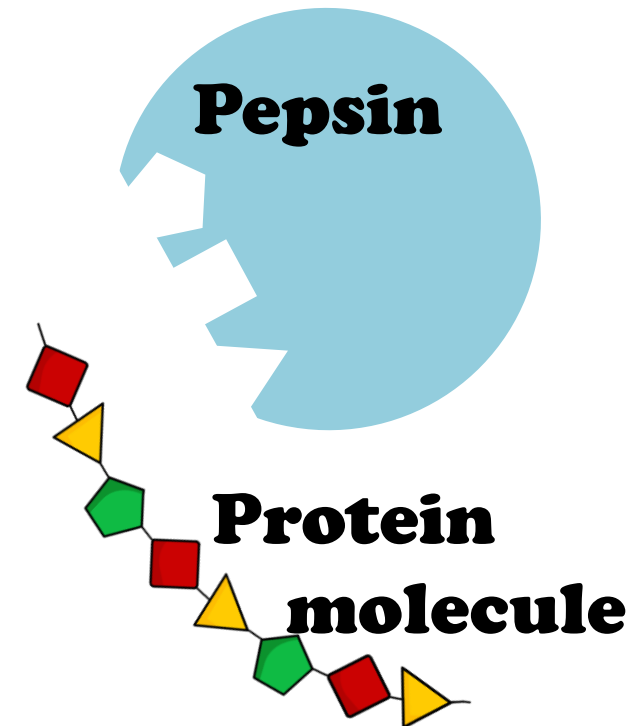
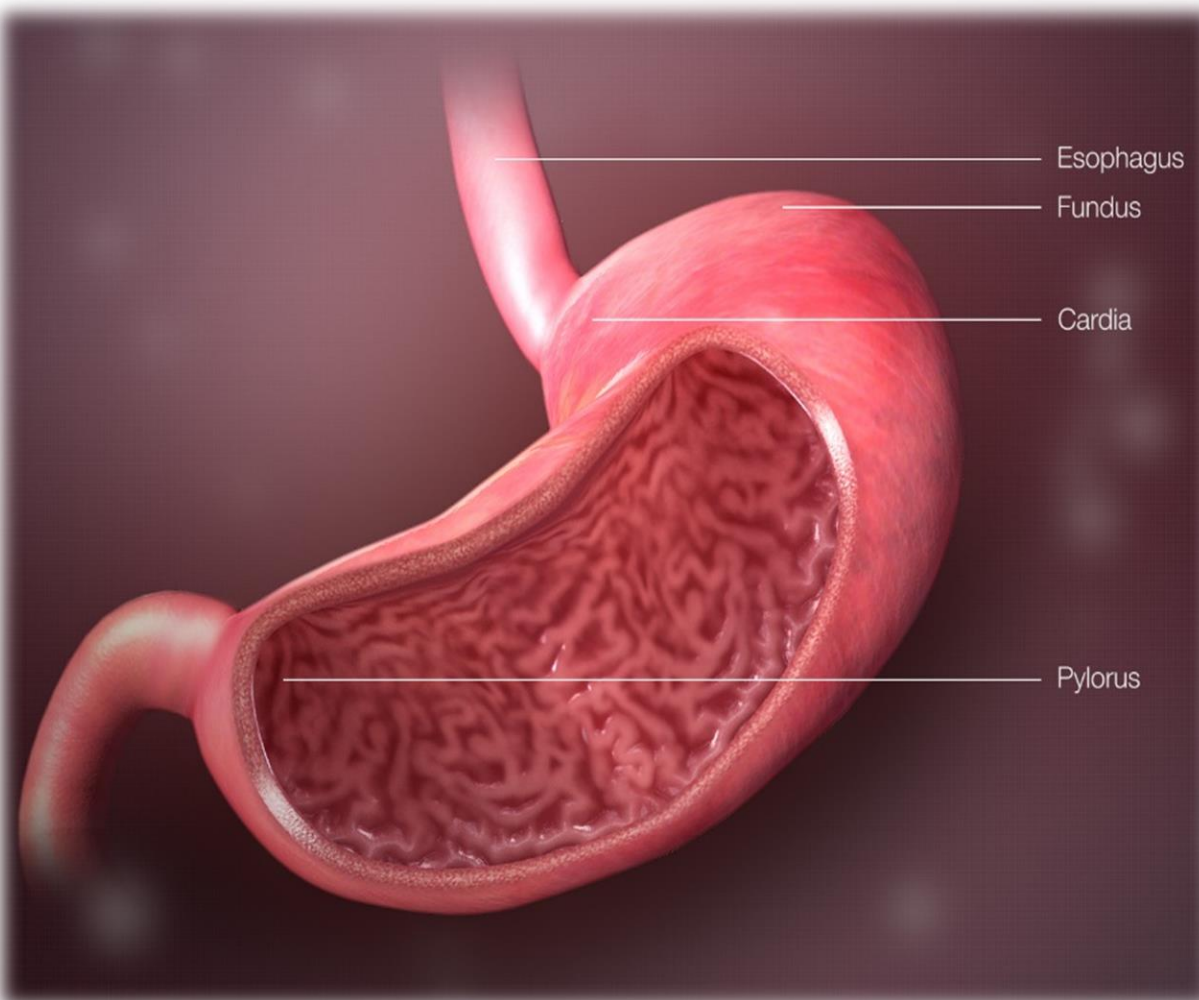
This graph shows us rates of reaction of different enzymes in the digestive system.

What is the substrate for pepsin? → What are its optimal conditions?

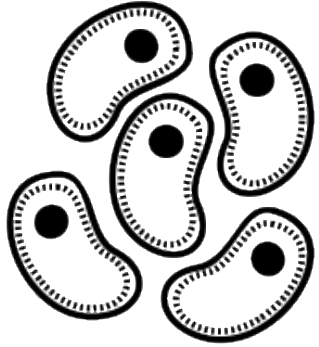
Pepsin digests protein and from the graph pH 1 is optimal



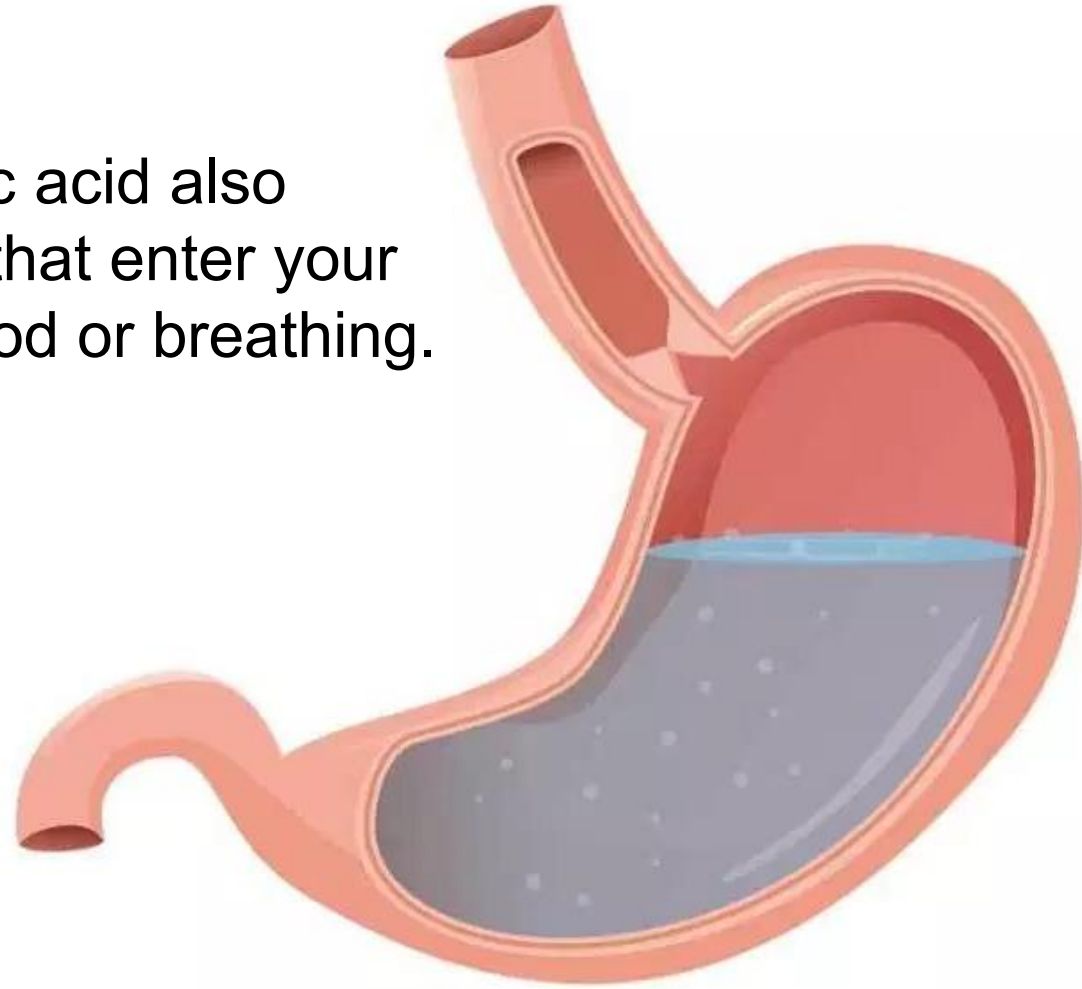
Pepsin is a **protease enzyme** secreted from glands.
Its **optimal pH is acidic (1-2 pH)**. The stomach releases hydrochloric acid from the same glands.



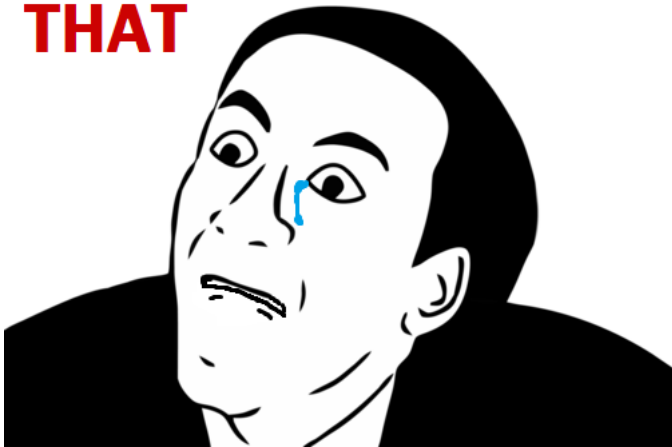
Stomach acid creates an **optimal environment** for enzymes to work.



The hydrochloric acid also kills pathogens that enter your body through food or breathing.



**PLEASE DON'T SAY
THAT**



~~The acid digests the food.~~

This table shows the mass of meat left after being placed into different test tubes of solution for 2 hours. The same mass was put into each test tube. (see pic)

Meat	Pepsin	HCl	Pepsin + HCl
Steak	1.3g	1.5g	0g

1. What conclusions can be made from the data
2. The mass of meat never changed when placed in HCl. Explain why.
3. Why is HCl and pepsin the most effective at Digesting the meat?

Review

Meat	Pepsin	HCl	Pepsin + HCl
Steak	1.3g	1.5g	0g

1. What conclusions can be made from the data

Pepsin + HCl was the most effective at digesting as there was no meat left. Pepsin does digest but only slightly and HCl doesn't digest at all.

2. The mass of meat never changed when placed in HCl. Explain why. **Acid can't digest meat**

3. Why is HCl and pepsin the most effective at Digesting the meat?

Acid creates the optimal conditions for the enzyme to work.

Your stomach has a **mucus layer** than **protects** your **stomach** **from being digested** by the pepsin and being damaged by the acid. If the layer gets damaged you can get a painful **ulcer**.



H. Pylori

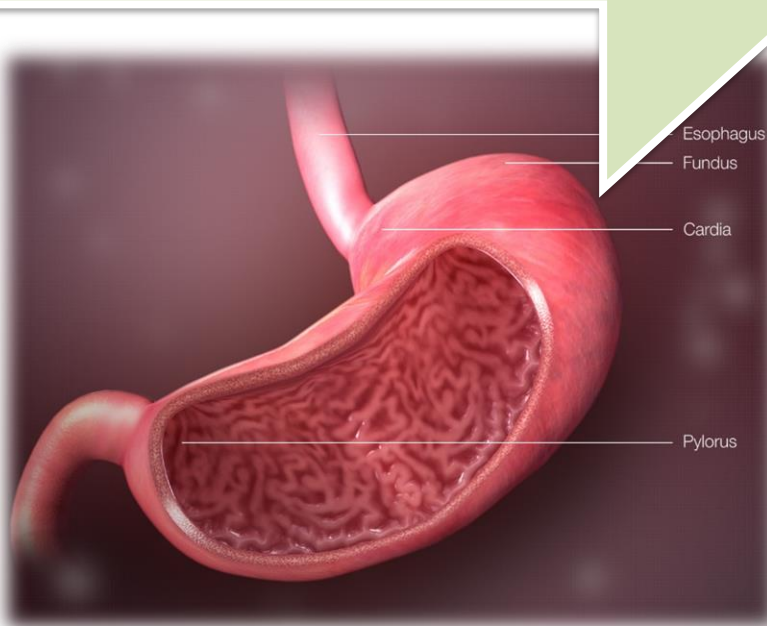
Damage to the lining is normally caused by bacteria that don't get killed by the acid. They damage the lining and the stomach gets digested by its own acid and enzymes.

LOb: Understand the role of the stomach and liver in producing substances that increase efficiency of digestion.

Keywords: Carbohydrase, amylase, protease, amino acid, lipase, glycerol, fatty acid, soluble, absorb, emulsification

Learning Outcomes:

- ✓ Describe how the pH is changed in the digestive tract to improve efficiency
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Did you know... the stomach has about 35 million glands in its lining. They secrete important substances to aid digestion.

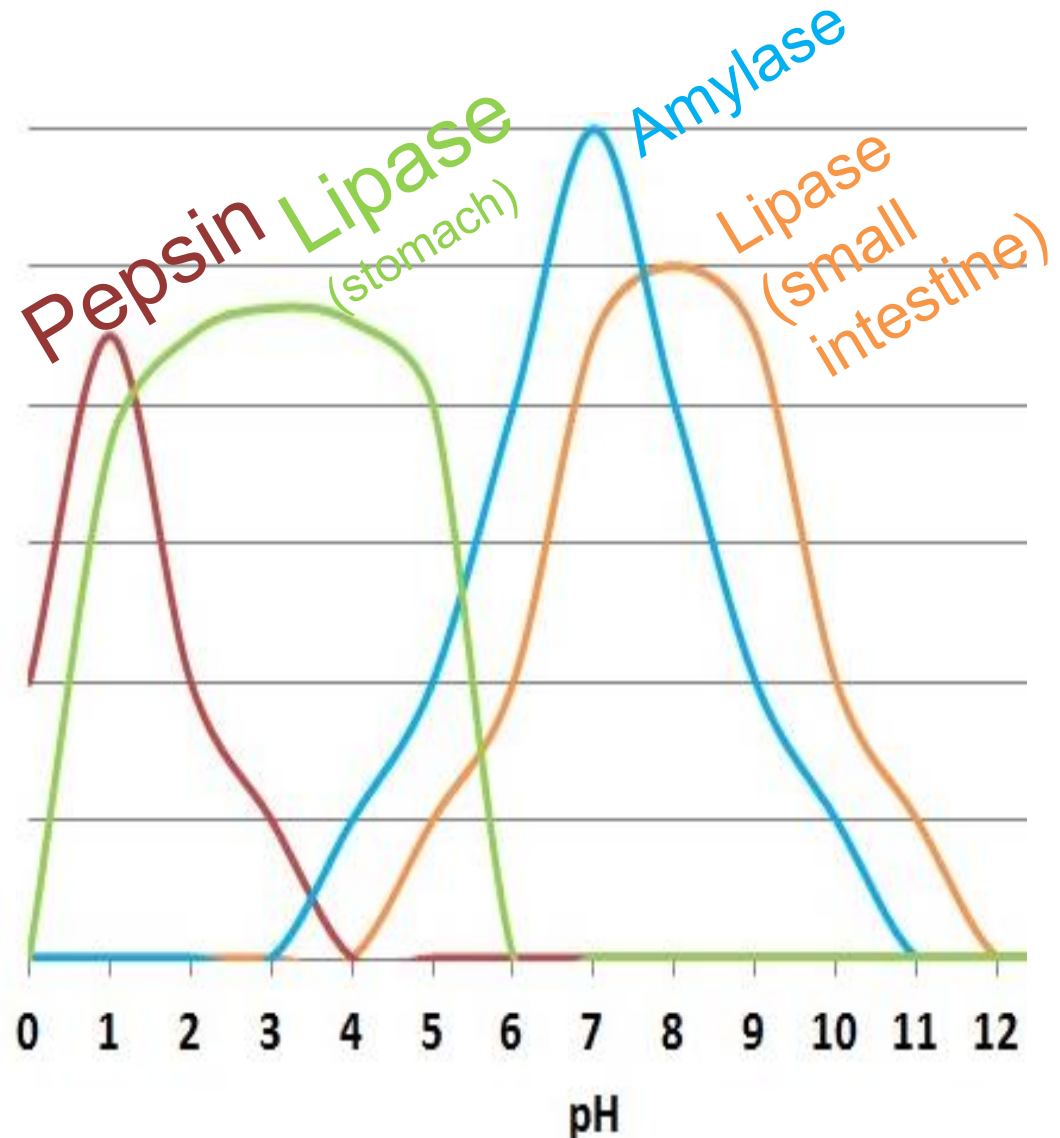
There is a problem when the semi digested food moves from the stomach to small intestine.

pH2

The food coming into the small intestine is acidic.

The enzymes in the small intestine work best in alkali conditions

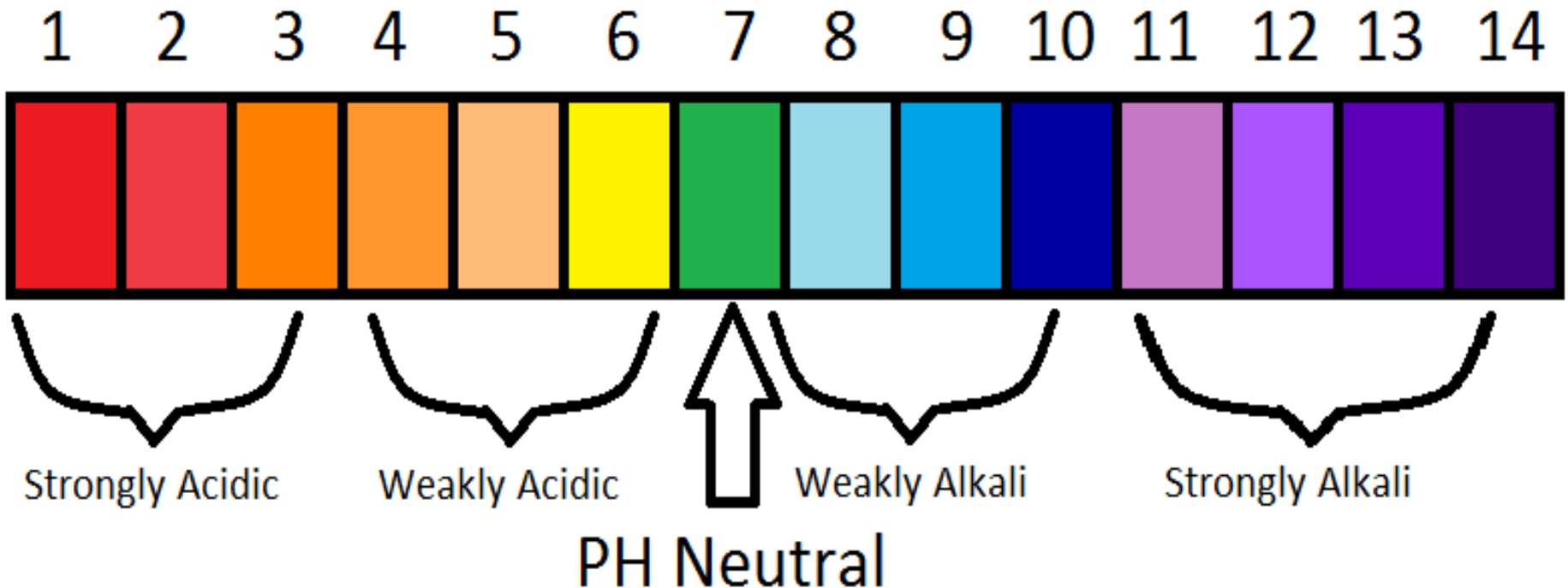
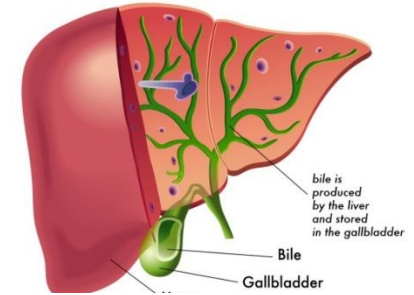
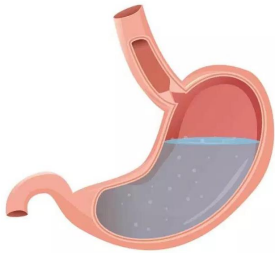
pH8



The acid needs to be neutralised

Neutralisation: Means to cancel out

The **liver** produces a substance called **bile** that **neutralises the stomach acid** to increase the pH.

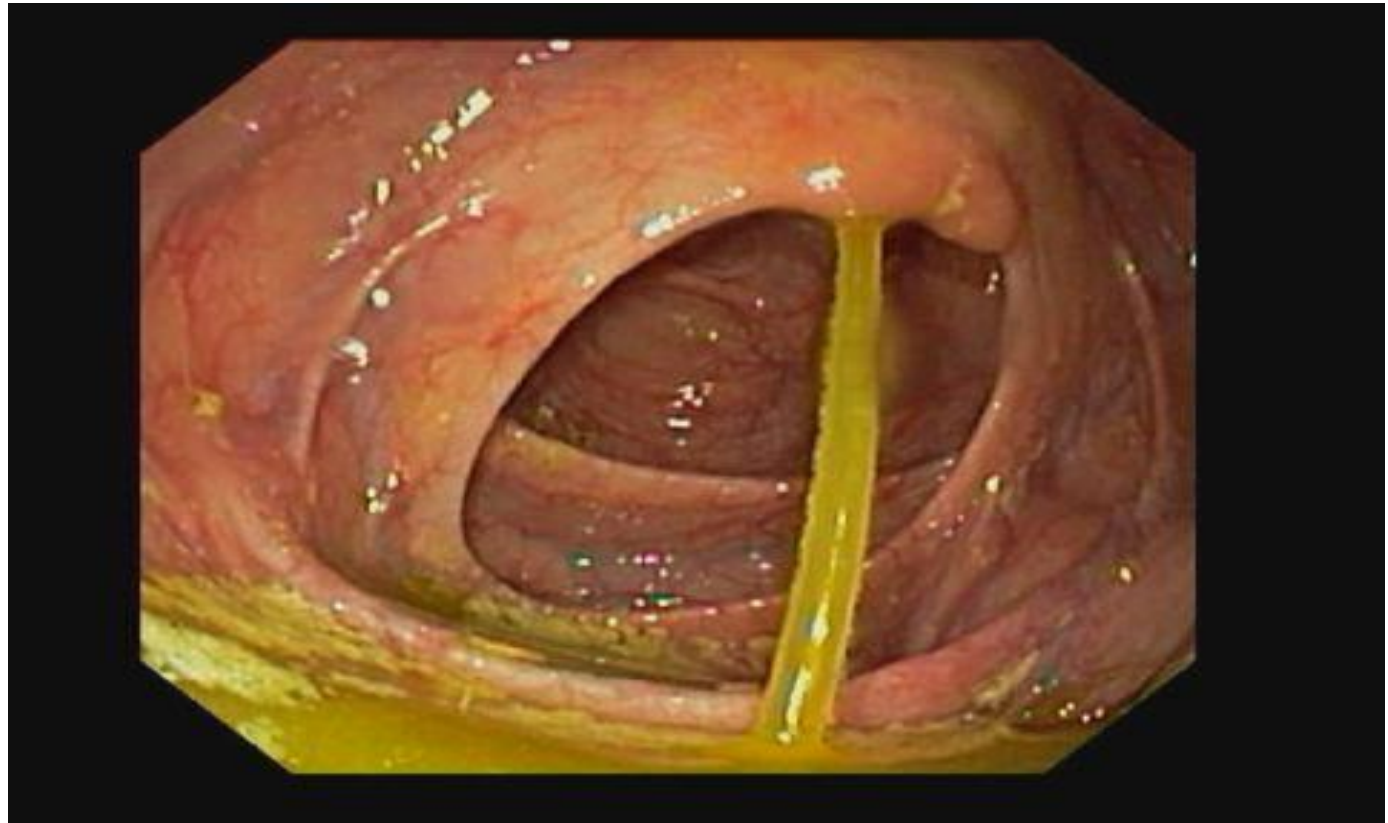
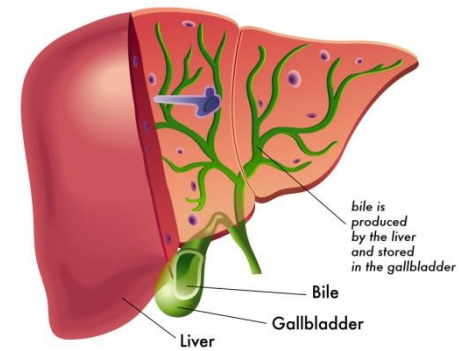


Bile

A green liquid **made in the liver**, stored in the **gall bladder** and **released into the small intestine**.

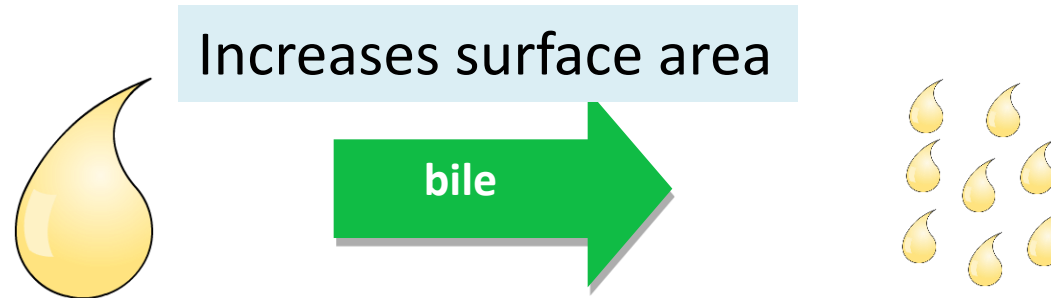


Neutralises stomach acid so the acid does not damage the small intestine.

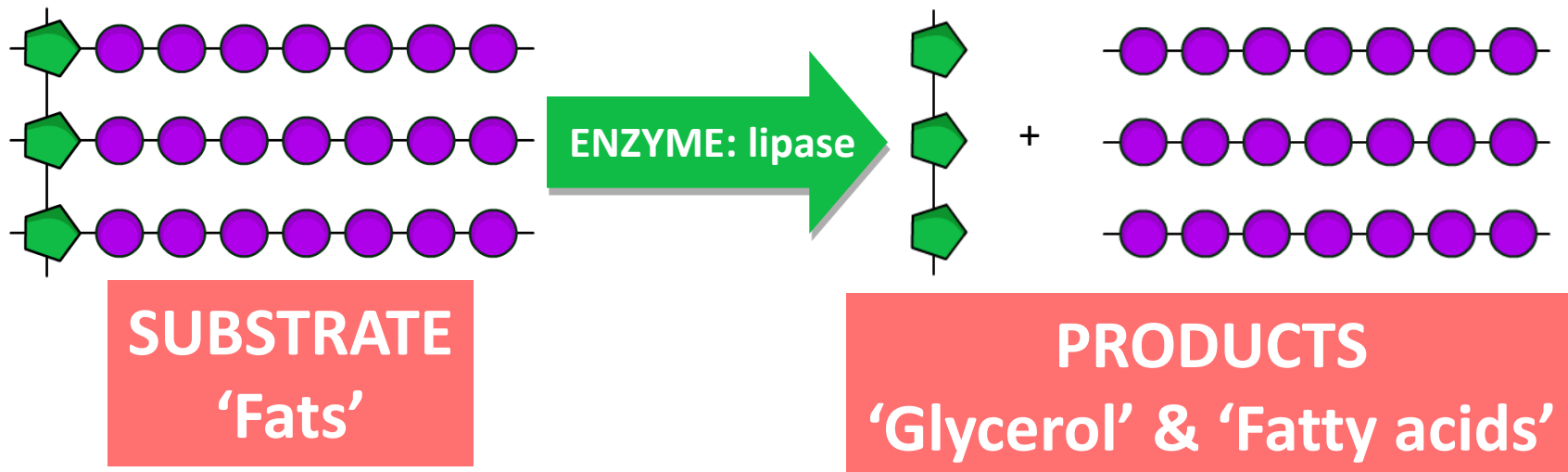


Bile improves the efficiency of fat digestion.

1. **Bile** (NOT an enzyme) allows the fat in our food to 'mix' with water by breaking the fat into smaller droplets. This is called **emulsification**.



2. **Lipase** working in an optimal pH environment provided by the bile breaks down each fat molecule → glycerol and fatty acid molecules.



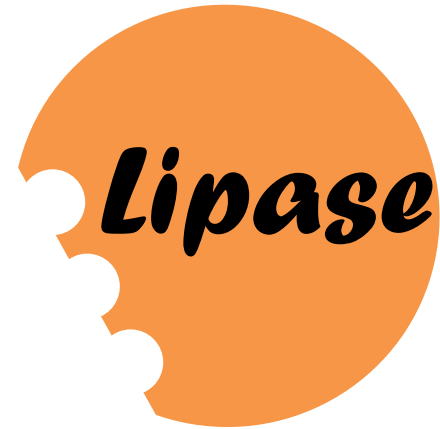
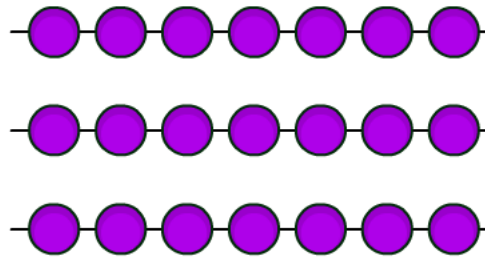
Fatty acids are acidic. As lipase digests the fat the pH will decrease.

Glycerol
molecule



+

Fatty acid
molecule

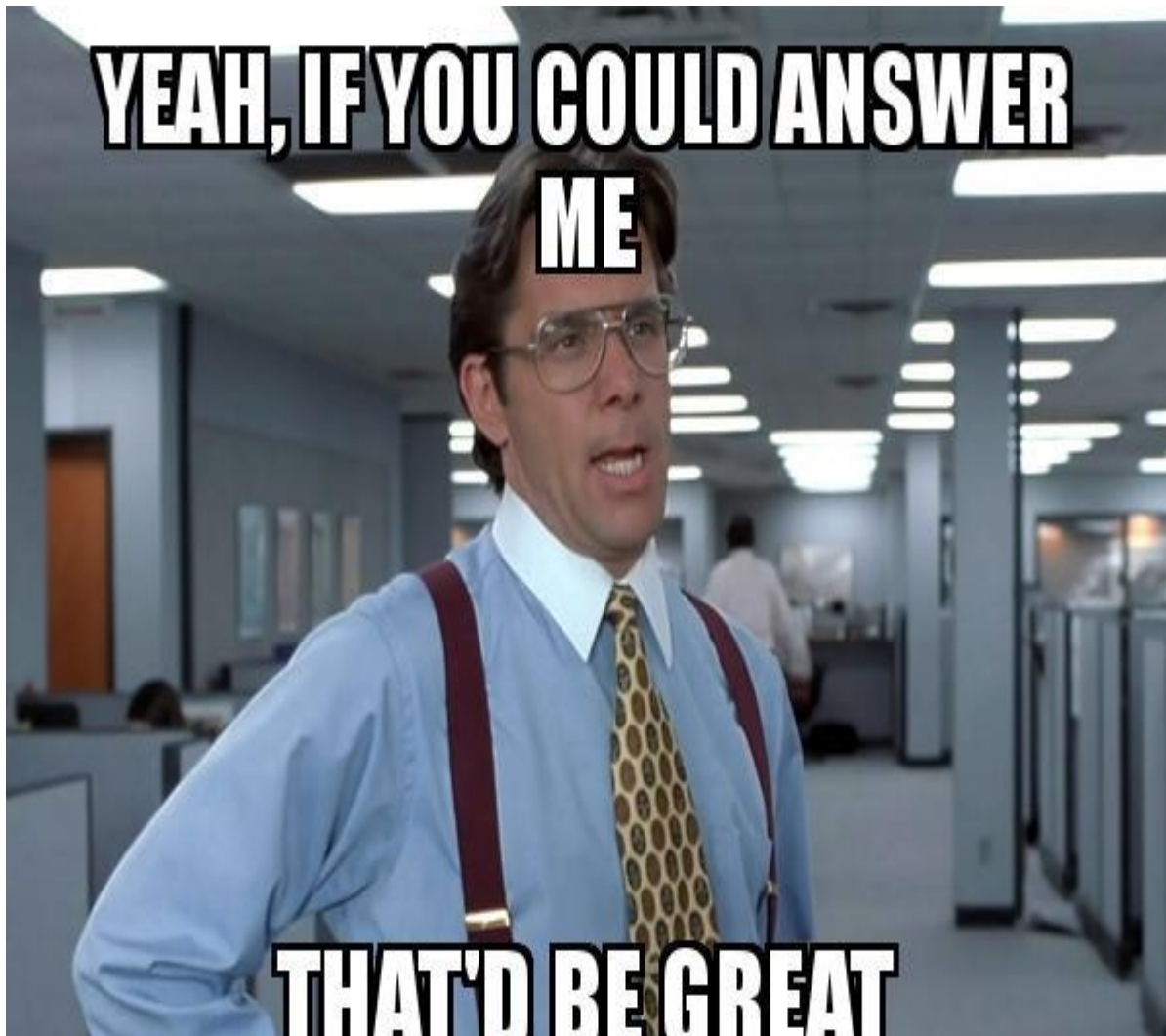


Slight decrease in pH caused
by fatty acid production

1 2 3 4 5 6 7 8 9 10 11 12 13 14

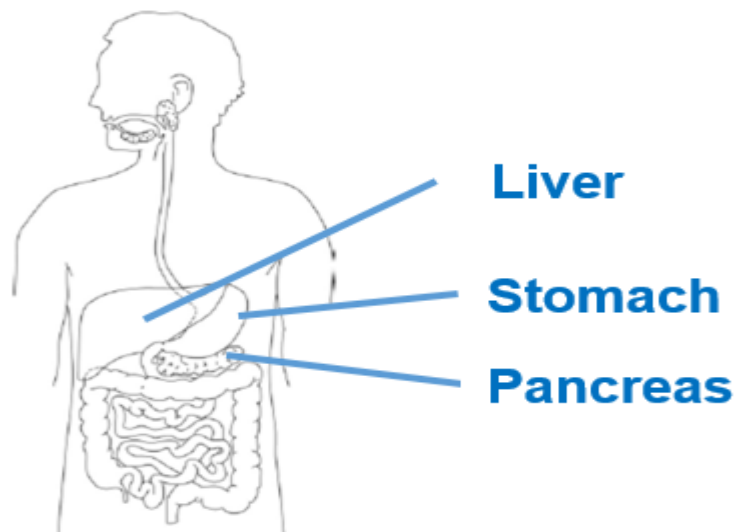


Activity: Answer the exam style questions on your sheet



Questions

1 a Label the liver, stomach and pancreas on this diagram.



(2 marks)

b People can suffer from stomach ulcers caused by a species of bacteria called *H. Pylori*. Describe how an infection by these bacteria might cause a stomach ulcer.

Bacteria no killed by the stomach acid so they damage the lining [1]

So the acid (HCl) damages the stomach tissue / causes an ulcer [1]

(2 mark)

b Enzymes also damage the lining causing an ulcer. Name an enzyme made by the stomach.

..... **Pepsin**

(1 mark)

2 Answer these questions:

a Name a substance, produced by the liver, involved with digestion?

Bile

(1 mark)

2 Answer these questions:

a Name a substance, produced by the liver, involved with digestion?

Bile

.....

(1 mark)

b Where is this substance stored?

Gall bladder

.....

(1 mark)

c This substance is involved in emulsification of fats. Explain this.

Breaks down fat into smaller droplets (allows mixing with water) [1]

This increases the surface area of fat for easier digestion [1]

(2 mark)

d What does this substance do to the pH of food coming from the stomach?

..... **Increases the pH**

(1 mark)

3 Some students investigated the effects of bile on the digestion of fat by lipase:

a The students:

1. mixed milk and bile in a beaker
2. put the pH sensor of a pH meter into the beaker
3. added lipase solution
4. recorded the pH at 2-minute intervals
5. repeated steps 1 to 4, but used water instead of bile.

Suggest two variables the students should have controlled

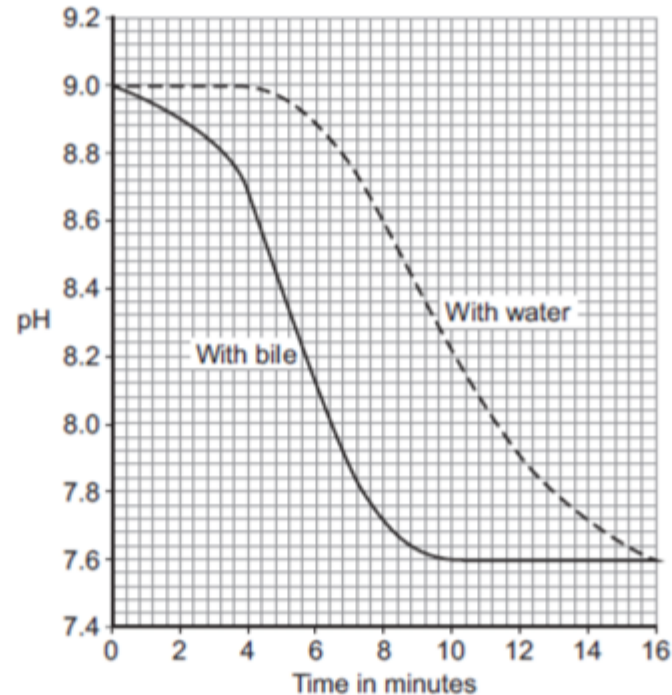
1. **Any two:**
Type of milk [1] Equal volume of bile to water [1]
2. **Volume of milk [1] Volume of lipase [1]**
Concentration of lipase [1] Temperature [1]

(2 mark)

b Below is a graph of their results. Why did the pH decrease in both investigations?

.. **Fatty acid (production)**

(1 mark)



c Bile helps with the digestion of fat. What evidence is there for this from the graph?

Fast reaction / digestion with bile [1] Steeper line with bile [1]

pH decreases faster with bile [1] Less time with bile [1]

(1 mark)

d Why did the pH decrease to the same level in both tests?

All the fat is digested [1]

Some amount of fatty acids [1]

Lower pH denatures the enzymes [1]

(1 mark)

Stomach

- Produces hydrochloric acid
- Reduces pH (pH1-2)
- Optimal environment for enzymes to work
- Pepsin optimal pH is pH 1

Liver

- Produces bile
- Stored in gall bladder released into small intestine
- Bile is alkaline so increases pH of food from stomach
- Bile makes lipase enzyme work more efficiently in small intestine as its optimal pH is between 8-9 pH
- Bile emulsifies fats
- Break down fat into small droplets
- Increases surface area of fat for lipase digestion

Level 3: Relevant points (scientific knowledge) are identified, given in detail and linked logically to form a clear account. [5-6 marks]

(6 mark)

Level 2: Relevant points (scientific knowledge) are identified and there are attempts at logical thinking. The resulting account is not fully clear. [3-4 marks]

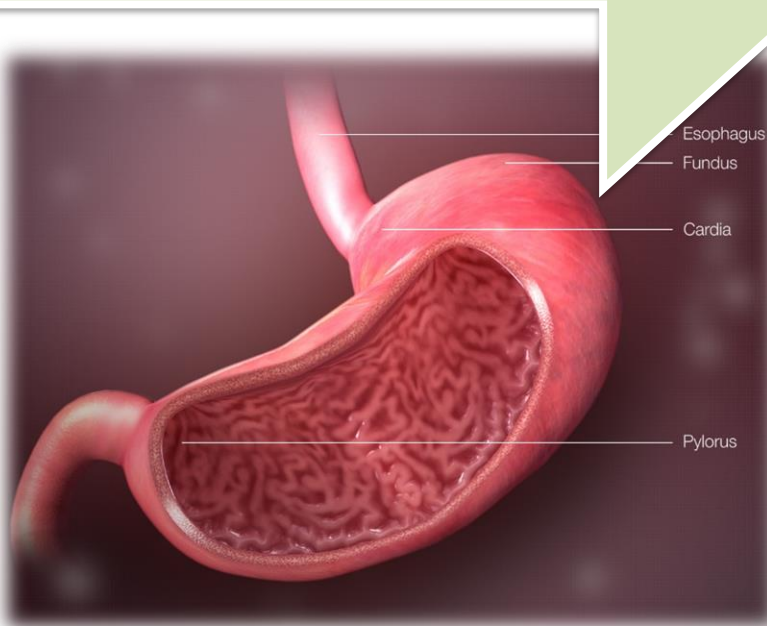
Level 1: Points are identified and stated simply, but their relevance is not clear and there is no attempt at logical thinking. [1-2 marks]

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Did you know... the stomach has about 35 million glands in its lining. They secrete important substances to aid digestion.

keyfacts



Stomach produces **hydrochloric acid** which has two functions
Decrease pH for **enzyme to work efficiently** and **kills pathogens**.



Bile is alkaline. It is **produce** by the **liver**, **stored** in the **gall bladder** and **released** into the **small intestine**.



Bile increases the pH **neutralising stomach acid**. Provides optimal pH for enzymes produced by the **pancreas** and **small intestine** to work. They like **alkaline conditions**.



Bile emulsifies fats. Allows them to mix with water. **Increases the surface area of fat** by breaking them into **small droplets**.

Plenary

Write three bullet points. Something you learnt at the beginning, middle and end today.

